The reduction of OEIS A396102 modulo 3

Adrian Perez Fontelles
Independent researcher

31 August 2026

Abstract

OEIS A396102 is defined by a functional equation for its ordinary generating function and carries a congruence conjecture of P. D. Hanna. It is true. The equation determines the coefficients one at a time, with $a(N)$ entering with coefficient 1, so it determines them modulo 3 as well; and modulo 3 the series $x/(1-x)$ satisfies the equation exactly, because its compositional iterates are $x/(1-jx)$ and the relevant product of denominators collapses. Uniqueness then identifies the reduction, which is the conjecture. The whole argument is an identity between rational functions over $\mathbb{Z}/3\mathbb{Z}$; nothing is truncated or estimated.

2020 Mathematics Subject Classification. 11B50, 05A15, 13F25.

1 The sequence and the conjecture

OEIS A396102 is "G.f. A(x) satisfies A(A(A(x))) = (1+x) * A(A(x)).". It has offset 1 and begins

$$1, 1, -2, 10, -71, 616, \ \ldots$$

Writing $A(x) = \sum_{n\geq 1} a(n)x^n$ and A^j for the j-th compositional iterate of A, the entry defines A by

$$A^3(x) \ = \ (1+x)\, A^2(x). \tag{1}$$

This is the only input taken from the entry.

The entry separately carries the following comment:

> Conjecture: a(n) == 1 (mod 3) for n >= 1. — *Paul D. Hanna*

As of the "Last modified" line on the live entry (Jun 08 2026 10:53:10, revision 14) it is still recorded as a conjecture, and nothing on the entry records it as settled either way.

2 The equation determines the sequence

Lemma 1. *Let $A(x) = \sum_{n\geq 1} a(n)x^n$ with $a(1) = 1$. Then for every $j \geq 1$ and every $N \geq 1$,*

$$\left[x^N\right] A^j = j\, a(N) + \big(\text{a polynomial in } a(1), \ldots, a(N-1)\big).$$

Proof. Induction on j. For $j = 1$ the statement is $\left[x^N\right] A = a(N)$. Suppose it holds for j, and write $B = A^j$, so $\left[x^1\right] B = 1$ and $\left[x^N\right] B = j\, a(N) + \cdots$. Then $A^{j+1} = A \circ B = \sum_{m\geq 1} a(m) B^m$ and

$$\left[x^N\right] A^{j+1} = \sum_{m=1}^{N} a(m) \left[x^N\right] B^m.$$

The term $m = N$ contributes $a(N)\big(\left[x^1\right] B\big)^N = a(N)$. The term $m = 1$ contributes $\left[x^N\right] B = j\, a(N) + \cdots$. For $2 \leq m \leq N-1$, every monomial of $\left[x^N\right] B^m$ is a product of $m \geq 2$ coefficients of B with indices summing to N, hence each index is at most $N-1$, so no $a(N)$ occurs. Adding up gives $(j+1)a(N)$ plus terms in $a(1), \ldots, a(N-1)$. $\square$

Lemma 2. *There is exactly one sequence of integers $a(1), a(2), \dots$ satisfying (1), and for each $N \geq 2$ the equation read at order x^N has the form $a(N) = ($a polynomial with integer coefficients in $a(1), \dots, a(N-1))$.*

Proof. Comparing coefficients of x in (1) gives $a(1) = 1$. Let $N \geq 2$ and collect the coefficient of $a(N)$ on each side of (1) at order x^N, using Lemma 1. On the left, $[x^N]A^3 = 3a(N) + \cdots$. On the right, $[x^N]\big((1+x)A^2\big) = [x^N]A^2 + [x^{N-1}]A^2 = 2a(N) + \cdots$, the second summand involving only $a(1), \dots, a(N-1)$. Hence the net coefficient of $a(N)$ is $3 - 2 = 1$. Since that coefficient is 1, the order-x^N equation solves for $a(N)$ over $\mathbb{Z}$, and induction gives existence and uniqueness. $\square$

Corollary 3. *Let $m \geq 2$. If $\bar{A} \in (\mathbb{Z}/m\mathbb{Z})[[x]]$ satisfies $\bar{A} = x + O(x^2)$ and (1) with all coefficients read modulo m, then $a(n) \equiv [x^n]\bar{A} \pmod{m}$ for every n.*

Proof. Reduction modulo m is a ring homomorphism, so it carries the integer recursion of Lemma 2 to the same recursion over $\mathbb{Z}/m\mathbb{Z}$, where the coefficient of $a(N)$ is still 1 and hence still invertible. That recursion therefore has a unique solution, and both $a \bmod m$ and the coefficient sequence of $\bar{A}$ solve it. $\square$

3 The reduction modulo 3

Write $f(x) = \dfrac{x}{1-x}$ over $\mathbb{Z}/3\mathbb{Z}$.

Lemma 4. $f^j(x) = \dfrac{x}{1-jx}$ *for every $j \geq 1$.*

Proof. For $j = 1$ this is the definition, and if $f^j(x) = x/(1-jx)$ then

$$f^{j+1}(x) = \frac{\dfrac{x}{1-jx}}{1 - \dfrac{x}{1-jx}} = \frac{x}{(1-jx) - x} = \frac{x}{1-(j+1)x}.$$

$\square$

Lemma 5. *Over $\mathbb{Z}/3\mathbb{Z}$, f satisfies $f^3 = (1+x)f^2$.*

Proof. By Lemma 4, $f^3(x) = \dfrac{x}{1-3x}$, and $3 \equiv 0$ modulo 3, so $f^3(x) = x$. Again by Lemma 4, $f^2(x) = \dfrac{x}{1-2x}$, and $-2 \equiv 1$ modulo 3, so $f^2(x) = \dfrac{x}{1+x}$. Therefore

$$(1+x)\, f^2(x) = (1+x) \cdot \frac{x}{1+x} = x = f^3(x).$$

$\square$

Theorem 6. *Over $\mathbb{Z}/3\mathbb{Z}$ the reduction of A is $x/(1-x)$; equivalently $a(n) \equiv 1 \pmod{3}$ for every $n \geq 1$. This is the conjecture of Section 1.*

Proof. By Lemma 5, $f = x/(1-x)$ satisfies the reduced equation, and $f = x + O(x^2)$. By Corollary 3 the reduction of A equals f, whose every coefficient is 1. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the recursion of Lemma 2 was run in exact rational arithmetic and its output compared against the entry's DATA at every one of the 23 published indices; the values agree exactly, and every one is an integer. A recursion that reproduced only some of them would mean the functional equation had been transcribed wrongly.

Second, at every step the coefficient of $a(N)$ was computed rather than assumed, by evaluating the order-x^N equation at $a(N) = 0$ and at $a(N) = 1$ and subtracting; it came out 1 at every N up to 25, as Lemma 2 requires. The same run was continued to $n = 25$ and every term reduced modulo 3; all residues are 1.

Third, the composition of the proof was carried out independently as a formal power series over $\mathbb{Z}/3\mathbb{Z}$ to order 30, starting from the coefficients of $x/(1-x)$ rather than from the closed form; both sides of (1) agree at every order, and the iterates f^j were confirmed to have coefficients j^{n-1} for $j \leq 6$.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A396102.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999.

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.